

PROBLEM CONTEXT

Fraudsters distribute clone banking applications via bulk SMS and WhatsApp, targeting India's 750 million smartphone users. Sideloaded APKs harvest credentials, intercept OTPs through overlay attacks, or embed Remote Access Trojans granting full device control. Existing defences — CERT-In blacklisting, TRAI sender filters, Play Protect — are reactive, with a 48–72 hour response lag. A single campaign reaches lakhs of recipients before any block is enforced. No current mechanism intercepts a phishing link before it reaches the victim's device, nor correlates fraud signals across institutions in real time.

CORE IDEA

SurakshaSetu is a three-layer federated interception network where each layer independently stops the attack — and all three operate in concert so that evasion of one layer does not result in financial loss. Protection is enforced simultaneously at the **telecom gateway** (before delivery), on the **device** (before installation), and inside the **Core Banking System** (before funds move).

ARCHITECTURE OVERVIEW

Layer 1 — Network Edge	Hooks into the SMSC/DLT gateway. Every URL in an inbound SMS or WhatsApp Business message is scored against a live Threat Registry before delivery. High-risk links replaced with a warning redirect in under 800 ms. Integration: TRAI DLT API + telecom SMSC hooks (Jio, Airtel, BSNL, Vi).
Layer 2 — Device Agent SDK	Lightweight SDK embedded within SBI YONO (under 6 MB, Android Go compatible). Monitors APK installation events, fingerprints each package's permission-request graph, and detects zero-day variants without signature updates. On alert: full-screen warning served; hash pushed to registry within 90 s.
Layer 3 — Core Banking	Device-risk signals ingested directly into the CBS transaction engine. Flagged devices trigger out-of-band IVRS voice verification — bypassing SMS OTP, which a compromised device can intercept. Integration: SBI CBS API + NPCI UPI transaction hooks.



Fig. 1 — Three-layer interception: network edge (L1), device agent (L2), core banking (L3). Dashed lines indicate cross-layer risk signal propagation into the Federated Threat Registry.

KEY FEATURES

Pre-installation App Authenticity Check

APK SHA-256 hash and permission-graph fingerprint validated against the registry before installation completes. Zero-day variants caught without signature updates.

URL / Link Scanning Before Click

Telecom-level interception scores every inbound link; warning redirect in under 800 ms. Covers SMS and WhatsApp Business API channels.

Device-Level Warning System

Full-screen alert with block / proceed options. All responses logged for cross-institution intelligence.

Central Fraud Intelligence Registry

Quorum-validated, hash-only federated ledger. Three-member consensus promotes threat entries to the active blocklist within 15 minutes. No raw PII shared — DPDP Act compliant.

OOB IVRS Authentication

Out-of-band voice verification for flagged devices — eliminates SMS OTP interception on compromised handsets.

WhatsApp Business API Integration

Direct pipeline from India's primary fraud distribution channel into the registry. User-reported malicious links trigger registry updates in real time.

USER JOURNEY — MALICIOUS LINK TO PREVENTION

01	Link Received Victim receives SMS or WhatsApp message containing a malicious URL.	NETWORK
02	SMSC Interception URL extracted and scored against the live Threat Registry before delivery.	
03	Block / Warn High-risk links replaced with a warning redirect. Low-risk links pass through.	BLOCKED
04	APK Sideload Attempt If user proceeds, Device Agent SDK detects the sideload event.	
05	Fingerprint Check APK hash and permission-graph matched against known malware signatures.	DEVICE
06	On-Device Alert Full-screen warning served. Block option presented. Hash pushed to registry.	
07	Transaction Attempt If device is flagged, CBS intercepts the incoming transaction request.	CBS
08	OOB Verification CBS triggers IVRS voice call — out-of-band authentication, bypassing SMS OTP.	
09	Clearance / Block Verified transactions proceed; unverified held and fraud team alerted.	CLEARED

UNIQUE SELLING PROPOSITION

No existing Indian solution combines telecom-level interception + on-device fingerprinting + CBS-integrated OOB authentication in a single coordinated framework.

Three-barrier defence: An attacker must independently defeat all three layers. Evasion of L1 does not bypass L2; evasion of L2 does not bypass L3 — architecturally unlike any point solution currently deployed.

Permission-graph fingerprinting detects zero-day APK variants without signature database updates, closing the 48–72 hour window exploited by current campaigns.

Federated Threat Registry enables cross-institution intelligence sharing under DPDP Act data minimisation. No raw PII leaves any institution; only cryptographic hashes are shared.

SCALABILITY AND FEASIBILITY

Low-End Device Support	Agent SDK runs on Android Go (2 GB RAM), under 6 MB footprint — the exact handset tier exploited by attackers.
Telecom Scale	URL scoring API integrates via TRAI DLT hooks available to all licensed operators. Deployable across Jio, Airtel, BSNL, Vi without custom hardware.
Minimal User Friction	Warning redirect and on-device alert require a single tap. No additional app installation beyond the existing YONO update cycle.
Phased Rollout	Phase 1 (0–6 months): YONO SDK + Registry pilot within RBI Regulatory Sandbox. Phase 2 (6–12 months): SMSC integration with two operators + PSB onboarding. No new legislation required.
Regulatory Readiness	Deployable under existing RBI Cybersecurity Framework, TRAI DLT mandate, and CERT-In CII obligations.

TARGET IMPACT

<8 min Threat-to-block time	85% SMS/WA links intercepted	<5% Post-install fraud rate	0.2% False positive rate
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85% of malicious links blocked at the telecom gateway before victim interaction.

Post-install fraud rate reduced below 5% through device-agent interception and OOB authentication.

Threat-to-block latency reduced from 48–72 hours to under 8 minutes via registry consensus.

False positive rate held below 0.2% through hash-exact matching and three-member quorum validation.

Cross-institution intelligence shared across all participating PSBs within each 15-minute consensus cycle.

Domain	Regulatory	Deployment	Sandbox
Cybersecurity / Digital Banking	RBI TRAI CERT-In DPDP 2023	Production-ready: 6–12 months	Phase 1 within RBI Regulatory Sandbox